APPENDIX A

# Solvent Extraction Separations

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Solvent extraction is a technique which has been highly developed within many national nuclear energy programs because of its suitability as a selective separation process for fission products, actinides, and other radioactive substances. The technique is briefly described in several sections of this book (Sections 14.2.4.3, 14.3.4, 17.7.3.4, 18.2.6, 18.4.3, 21.7, and especially in 21.6 on the Purex process). It is based on the formation of uncharged organic metal complexes which are preferentially soluble in organic solvents. The three main types of such compounds are:

1. Organic chelate (i.e. binding through two positions in the organic molecule, like a claw) complexes such as plutonium tetraacetylacetonate, $PuAa_4$ (general formula $MA_n$) (Section 18.4.3, cf. Figs 18.3, 18.9 and 18.11);
2. Inorganic metal complexes forming adducts with solvating organic compounds like TBP and hexone, e.g. $UO_2(NO_3)_2)(TBP)_2$ (or generally $ML_nS_s$, where L is an inorganic anion and S is the organic adduct former (Section 21.6, cf. Fig. 14.9);
3. Ion pair complexes between large organic cations (like alkylammonium ions, $R_3NH^+$) and negatively charge inorganic complexes like $UO_2(SO_4)^-$ (generally $RB^+ML^-$, Section 14.2.4.2).

The physical and chemical principles governing the formation of extractable metal complexes, the conclusions which can be drawn about the chemical system from solvent extraction studies, and analytical and industrial applications are described in several monographs, e.g. 'Principles and Practices of Solvent Extraction' (2004).

## A.1. SINGLE STAGE BATCH EXTRACTIONS

Let us consider a solute (e.g. uranium) which is distributed between an organic and an aqueous phase, independent of the kind of compounds the solute forms in the two

phases. After equilibrium the weight of solute in the organic phase (*extract*) is $w_{org}$ and in the aqueous phase (*raffinate*) $w_{aq}$. Thus

$$w_{tot} = w_{org} + w_{aq} \tag{A.1}$$

The weight fraction in the organic phase is

$$\psi_1 = P/(P+1) \tag{A.2}$$

where the index 1 refers to the conditions after one extraction. $P$ is called the *extraction* (or *partition*) *factor* and defined by

$$P = D\theta \tag{A.3}$$

where $D$ is the distribution factor, defined by

$$D = \frac{\text{Concentration of all M-species in the organic phase}}{\text{Concentration of all M-species in the aqueous phase}} \tag{A.4}$$

and $\theta$ is the phase volume ratio $v_{org}/v_{aq}$. The percentage solute extracted, denoted $E(\%)$, is equal to $100\,\psi_1$. The fraction of nonextracted solute left in the organic phase is $\varphi_1$, and $\psi_1 + \varphi_1 = 1$,

$$\varphi_1 = 1/(P+1) \tag{A.5}$$

Assume that one wants to separate uranium from the fission product lanthanum using the two-phase system: aqueous 1 M $HNO_3$, the organic adduct former tributyl phosphate, TBP (100%, i.e. undiluted). In this system $D_U = 20$ for U(VI), while for lanthanum $D_{La} = 0.07$. If one extracts with a phase ratio of 0.5, then $\psi_U = 0.909$ ($E_U = 100\ \psi_U = 90.9\%$) and $\psi_{La} = 0.034$ ($E_{La}$ 3.4%). This may be unsatisfactory with respect to both uranium yield and purity.

The yield can be increased by repeated extractions of the same aqueous phase (multiple extraction with one stationary phase, or 'crosscurrent extraction'). For $n$ such extractions one finds that

$$\varphi_n = (P+1)^{-n} \tag{A.6}$$

Suppose $n = 3$ for our example, then $\varphi_{3,U} = 0.00075$, i.e. for the three organic phase volumes taken together $E_U = 100(1 - \varphi_{3,U}) = 99.92\%$. However, for lanthanum $\varphi_{3,La} = 0.902$, i.e. $E_{La} = 9.8\%$. Although the uranium yield is high, the lanthanum impurity may be intolerable.

A more elaborate technique must be employed in order to obtain both high yield and high purity under such conditions. Many such batch laboratory techniques have been described using alternatively fresh organic (*extraction*) and aqueous (*washing*) solutions. The extractions are carried out in special multistage equipment (Section A.4).

## A.2. MULTIPLE STAGE CONTINUOUS PROCESSES

A single partitioning of a compound between an organic solvent and water may not be sufficient for isolating it in acceptably pure form and good yield. Various multiple extraction techniques may therefore be required. Such techniques have been described in Section 21.6.4 and their technical application for uranium production (Section 14.2.4.3) and spent fuel reprocessing (Section 21.6.3).

Continuous processes are preferred in industry, where the most common and simple solvent extraction equipment is the mixer-settler (Fig. A.1(a); cf also Fig. 14.2). This type of equipment is also becoming standard in laboratories engaged in process development. In the uranium industry a single mixer-settler may hold as much as 1000 $m^3$. The mixer-settlers, each closely corresponding to a single ideal extraction stage, are arranged in

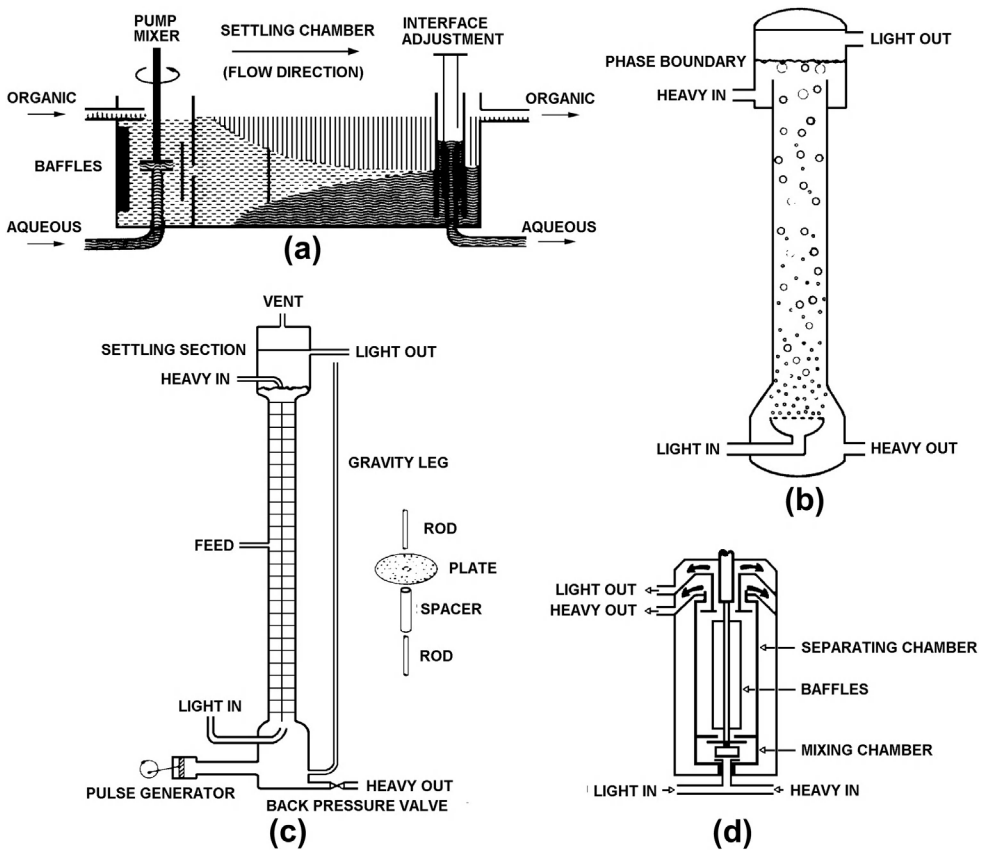



**Figure A.1** Different types of continuous extraction equipment. (a) Mixer-settler (b) Spray column (c) Pulsed column (d) Tubular centrifugal contactor.